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2027 CYCLE

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CUET UG

Chemistry

Haloalkanes and Haloarenes

Official Syllabus Unit VI · Class 12 · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus 2026, the current 2027 baseline, Unit VI)	Skills the questions test
Classification, naming and the C–X bond	Introduction, Classification of haloalkanes and Haloarenes, Nomenclature of Haloalkanes & Haloarenes, nature of C-X Bond	Sorting haloalkanes as 1°/2°/3°, allylic, benzylic, vinylic and aryl, writing IUPAC names, and predicting bond length/enthalpy trends down the halogens.
Preparation methods	Methods of preparation of Haloalkanes & Haloarenes	Choosing among HX, PX_3/PX_5 and $SOCl_2$ routes from alcohols, halide exchange (Finkelstein, Swarts), and arene routes (Sandmeyer, Balz-Schiemann, Gattermann).
Physical properties	Physical properties of Haloalkanes & Haloarenes (like Melting & boiling Point, Density and Solubility)	Ordering boiling points across halogens and isomers, predicting densities, and explaining water insolubility.
Reactions and stereochemistry	Chemical Properties of Haloalkanes and Haloarenes, Stereochemical aspects (optical activity, Chirality, retention, inversion, racemization etc) of S N- Reactions of Haloalkanes	Comparing S_N1 and S_N2 mechanisms, rate laws, inversion vs racemisation, eliminations (Saytzeff), metal reactions (Grignard, Wurtz, Fittig), and the stubbornness of aryl halides.
Polyhalogen compounds	Important Polyhalogen Compounds (CH_2Cl_2, $CHCl_3$, CHI_3, CCl_4, freons & DDT)	Naming each compound's use and hazard — from phosgene formation in chloroform to the ozone story of freons and the legacy of DDT.

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EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Chemistry paper maps strictly to NCERT Class 12, and Unit VI opens the organic half: it rewards crisp mechanism logic — S_N1 vs S_N2 — more than memorised equations.

Classification: One Family, Many Addresses

Halogenated organic compounds surround you — the chlorine sterilising your water supply, the iodine antiseptic in the clinic, the refrigerant gas in an old air-conditioner. **Haloalkanes** carry halogens on sp^3 carbons of an alkane framework; **haloarenes** attach halogens directly to an aromatic ring. Where exactly the halogen sits decides everything — reactivity, naming, even the medicine's safety — so the classification map below is the unit's real starting point.

- **By number of halogens:** mono-, di- and polyhalogen compounds —

  $$CH_3\ Cl$$

  versus

  $$CHCl_3$$

  versus DDT.
- **By the carbon carrying X (sp^3 family):**
  - **Primary (1°):** the carbon holds one other carbon —

    $$CH_3CH_2\ Cl$$

    .
  - **Secondary (2°):** two carbon neighbours — isopropyl chloride.
  - **Tertiary (3°):** three — tert-butyl chloride.
  - **Allylic:** X sits on the carbon **next to** a C=C ($CH_2 = CH - CH_2\ Cl$) — activated by the double bond.
  - **Benzylic:** X on the carbon next to a benzene ring (

    $$C_6H_5CH_2\ Cl$$

) — likewise activated.
- **By the carbon carrying X (sp^2 family):**
  - **Vinylic:** X directly on a C=C carbon ($CH_2 = CHCl$).
  - **Aryl (haloarene):** X directly on the ring carbon (

    $$C_6H_5\ Cl$$

    , chlorobenzene).
- **Why the split matters:** sp^3 carbons surrender their halogen easily (S_N chemistry galore); sp^2 carbons clutch theirs (vinylic and aryl halides refuse substitution — a whole section later explains why).

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Naming, in two minutes

- **IUPAC for haloalkanes:** halogen = fluoro/chloro/bromo/iodo prefix, alphabetical with other substituents; number the chain from the end nearest a substituent.

$$CH_3\ CH(Cl)CH_3$$

  = 2-chloropropane.
- **Haloarenes:** the ring positions are numbered 1–6; 1-bromo-4-chlorobenzene names a ring with Br at C1 and Cl at C4 (para to each other). Common names (o-, m-, p-) survive informally but exams lean IUPAC.
- **Dihalides with both halogens on one carbon** are geminal (gem-dihalides, from ketones + HX); on adjacent carbons, vicinal (from alkenes + X_2) — one-mark vocabulary.

The C–X Bond: Polar, Short, and Getting Weaker

- **Polarity:** halogens pull electron density harder than carbon (electronegativity F 4.0 > Cl 3.0 > Br 2.8 > I 2.5), so every C–X bond is polarised $C^{\delta+} - X^{\delta-}$. Yet haloalkanes stay insoluble in water — the polarity is too weak to pry water's hydrogen-bond network apart.
- **Bond length grows down the group** as the halogen atom swells: C–F (139 pm) < C–Cl (178) < C–Br (193) < C–I (214).
- **Bond enthalpy does the opposite** — bigger atoms share less effectively: C–F (452 kJ mol^{-1}) > C–Cl (351) > C–Br (293) > C–I (234). This single trend explains why **iodides react fastest**: the leaving bond snaps easiest.
- **The dipole moment surprise:** you would bet

$$CH_3\ F$$

  beats

$$CH_3\ Cl$$

  in dipole moment (F is more electronegative). Wrong: $CH_3Cl > CH_3F$, because fluorine's tiny size shortens the bond so much that (small distance × big charge) loses to chlorine's longer lever arm. A classic "which is larger" trap.
- **Reactivity preview:** weakest bond + best stabilised leaving ion makes the order of halide leaving ability F < Cl < Br < I — iodides are the sprinters.

Preparation: Building the C–X Bond

From alcohols — the workhorse route

- **Concentrated HX:**

$$R\text{–OH} + HX \rightarrow R\text{–X} + H_2\ O$$

  . Reactivity of HX: HI > HBr >> HCl; reactivity of ROH: 3° > 2° > 1° (the Lucas test — HCl + anhydrous

$$ZnCl_2$$

  — separates them by turbidity speed: 3° instantly, 2° in minutes, 1° only on heating).

- **Phosphorus halides:**

$$3R\text{–OH} + PCl_3 \rightarrow 3R\text{–Cl} + H_3PO_3$$

; red phosphorus + bromine/iodine gives the PBr_3/PBr_3 analogue; PCl_5 converts alcohols to chlorides too.

- **The cleanest route — thionyl chloride:**

$$R\text{–OH} + SOCl_2 \rightarrow R\text{–Cl} + SO_2 \rightarrow + HCl \rightarrow$$

. The by-products are both gases — they escape, leaving a **pure** alkyl halide. Exams love the phrase "preferred because the side products are gaseous".

From hydrocarbons

- **Free-radical halogenation:** Cl_2 or Br_2 under UV on alkanes gives mixtures of all possible isomers — cheap but unselective.
- **Electrophilic substitution on arenes:** benzene + Cl_2 over

$$FeCl_3$$

(anhydrous, dark) → chlorobenzene, with o/p isomers if a directing group already sits on the ring.

- **Sandmeyer reaction (from diazonium salts):**

$$C_6H_5N_2^+Cl^- + Cu_2Cl_2 \rightarrow C_6H_5\ Cl + N_2$$

— Cu_2Cl_2/Cu_2Br_2/CuCN in HX converts aryl diazonium salts to Ar–Cl, Ar–Br, Ar–CN with N_2 bubbling out.

- **Balz-Schiemann:**

$$ArN_2^+BF_4^-$$

heated → Ar–F — the standard route to aryl fluorides; **Gattermann** uses copper powder instead of cuprous salts.

- **Iodoarenes:** warm diazonium salt + potassium iodide — no catalyst needed.

Halide exchange — swap one halogen for another

- **Finkelstein:**

$$R\text{–X} + NaI \xrightarrow{\text{dry acetone}} R\text{–I} + NaX$$

— NaCl/NaBr precipitate out of dry acetone, pulling the equilibrium forward. Converts cheap chlorides/bromides into the more reactive iodides.

- **Swarts:** heating alkyl chlorides/bromides with metal fluorides (AgF$, $$Hg_2F_2$, CoF_2) gives fluoroalkanes — the route that built the freon industry.

Physical Properties: Mass Decides

- **Boiling points climb down the halogen column:** R–I > R–Br > R–Cl > R–F — heavier halogens mean more mass and stronger London dispersion forces. (Fluorides boil lower than the corresponding chlorides despite polarity — weak intermolecular attraction wins over dipole.)

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- **Boiling falls with branching:** among isomers, $1° > 2° > 3°$ — a branched molecule touches its neighbours less (smaller surface area).
- **Densities:** fluorides and chlorides float on water; bromides, iodides and polyhalides sink — and density **decreases** as the carbon chain grows (the heavy halogen gets diluted).
- **Solubility:** essentially insoluble in water (no hydrogen bonding possible), fully at home in organic solvents.
- **Melting point quirk:** the **para** isomer melts higher than the **ortho** — its symmetry packs the crystal lattice tighter. p-dichlorobenzene (mothballs) is the everyday proof.

Reactions of Haloalkanes: The Nucleophile Takes Over

The C–X bond's polarity leaves carbon electron-hungry — so electron-rich **nucleophiles** (OH^-, CN^-, NH_3, I^-) attack it, and the halide departs with the electron pair. Two mechanisms share this stage, and the exam's favourite question is telling them apart.

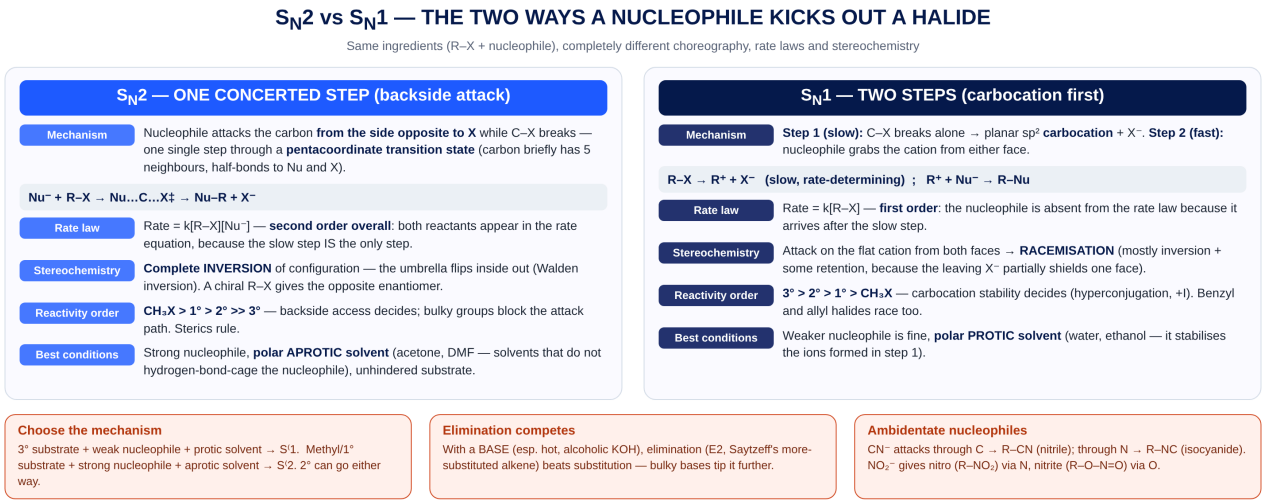


Figure 1: S_N2 versus S_N1 side by side: mechanism, rate law, stereochemical outcome, reactivity orders and best conditions for each.

S_N2: one concerted backside strike

- **The mechanism:** the nucleophile attacks the carbon from the side **opposite** to the leaving group; as C–Nu forms, C–X breaks — one step, no intermediates, a fleeting transition state where carbon is squeezed between both.
- **Kinetics:** both partners appear in the slow step, so Rate = k[RX][Nu⁻] — second order overall.
- **Stereochemistry:** the umbrella flips — **inversion of configuration** (Walden inversion). Chiral substrate in, opposite enantiomer out, 100%.
- **The reactivity order:** $CH_3X > 1^\circ > 2^\circ \gg 3^\circ$ — pure steric traffic: the backside road must be clear, and tertiary carbons are road-blocked.

S_N1: two steps with a carbocation

- **The mechanism:** C–X ionises **alone** (slow) into a planar sp^2 carbocation + X^-; the nucleophile then attacks whichever face it meets (fast).
- **Kinetics:** the nucleophile misses the slow step entirely — Rate = k[RX] only. First order.

- **Stereochemistry:** two faces, two products — **racemisation** (in fact slightly more inversion than retention, because the departing X^- lingers and shields its own side — a favourite nuance).
- **The reactivity order:** $3^\circ > 2^\circ > 1^\circ > CH_3X$ — carbocation stability rules (hyperconjugation and inductive support). Benzylic and allylic halides qualify too: their cations are resonance-stabilised.

The decisive comparison table

Feature	S_N2	S_N1
Steps	One, concerted	Two, via carbocation
Rate law	k[RX][Nu⁻] — second order	k[RX] — first order
Stereochemistry	Complete inversion	Racemisation (partial)
Substrate winner	Methyl > 1° > 2° >> 3°	3° > 2° > 1° (benzylic/allylic fast)
Nucleophile	Strong needed	Strength irrelevant (not in rate law)
Solvent	Polar aprotic (acetone, DMF)	Polar protic (water, ethanol)

Substitution's rivals: elimination and ambidentate nucleophiles

- **Elimination (dehydrohalogenation):** hot alcoholic KOH pulls HX out — a base removes a beta-hydrogen while X^- leaves, forging an alkene. **Saytzeff's rule:** the **more substituted** (more stable) alkene dominates — 2-bromobutane gives mostly but-2-ene, not but-1-ene. Elimination and substitution always compete; heat and bulky bases push toward elimination.
- **Ambidentate nucleophiles choose their teeth:** CN^- through carbon gives **nitriles** (R–CN, chain growth by one); through nitrogen gives **isocyanides** (R–NC). With KCN the carbon end (softer) attacks → R–CN; with AgCN the nitrogen end wins → R–NC. Same duel for NO_2^-: KNO_2 → nitrites (R–O–N=O), $Ag(NO_2)$ → nitro compounds (R–NO_2).
- **Aromatic wedding — Wurtz family reactions:**
  - **Grignard reagents:**

$$R\text{–X} + \text{Mg} \xrightarrow{\text{dry ether}} R\text{–MgX}$$

    — dryness is everything (water kills RMgX to RH). The carbon of RMgX behaves as a carbanion: the gateway to alcohols, acids, ketones in later chapters.
  - **Wurtz:** dry ether + Na couples two alkyl halides ($2R\text{–X} + 2\ \text{Na} \rightarrow R\text{–R} + 2\ \text{NaX}$) — good for symmetrical alkanes, messy otherwise.
  - **Fittig:** the same trick with two aryl halides; **Wurtz-Fittig** couples one of each (alkyl + aryl).

Haloarenes: Why the Ring Refuses

Chlorobenzene meets NaOH and… nothing happens at room temperature — the **same** hydroxide that converts alkyl chlorides in minutes. Aryl halides are the chapter's stubborn class, and four reasons stack up:

- **Resonance locks the bond:** the chlorine lone pair delocalises into the ring, giving C–Cl partial double-bond character — a much stronger, shorter bond (C–Cl: 169 pm in chlorobenzene vs 178 pm in alkyl chlorides).
- **sp^2 carbon grips harder:** the 33% s-character of the sp^2 C–Cl bond holds electrons closer and tighter than sp^3 does.
- **No friendly carbocation:** ionising would leave a **phenyl cation** — desperately unstable — so S_N1 is off the table.
- **Geometry blocks the backside:** the ring's π cloud repels the approaching nucleophile — S_N2 is off the table too.
- **The consequence:** hydrolysis needs brutal conditions — the **Dow process**: chlorobenzene + NaOH at 623 K under 300 atm, then acidification → phenol.
- **Nitro groups rescue the reaction:** electron-withdrawing NO_2 groups ortho/para to the halogen pull electron density from the reaction centre, enabling **nucleophilic aromatic substitution** (via the Meisenheimer complex) under mild conditions — 2,4,6-trinitrochlorobenzene hydrolyses almost easily.
- **Halogens are o,p-directing but deactivating:** resonance donates into the ring (directing o,p) while the -I effect withdraws (deactivating). Both statements are true at once — the exam's favourite pair of options.

Polyhalogen Compounds: Useful and Dangerous

- **Dichloromethane (**

$$\mathbf{CH_2Cl_2}$$

): paint stripper, metal degreaser, extraction solvent; moderately toxic — ventilate.
- **Trichloromethane (**

$$\mathbf{CHCl_3}$$

, chloroform): the historical anaesthetic (now retired — liver damage). Slowly oxidises in air/light to the deadly gas **phosgene** (

$$COCl_2$$

) — hence storage in dark blue bottles filled to the brim.
- **Tri-iodomethane (**

$$\mathbf{CHI_3}$$

, iodoform): yellow crystalline powder with an antiseptic smell — the old dressing-room classic (its iodine content does the work).
- **Tetrachloromethane (**

$$\mathbf{CCl_4}$$

): once a fire-extinguisher fluid and degreaser — but it produces phosgene on hot metal surfaces and wrecks the liver, so it is phased out.

- **Freons (CFCs, e.g.**

$$\mathbf{CFCl_3}$$

, freon-12

$$\mathbf{CF_2Cl_2}$$

): odourless, non-toxic, non-flammable — the perfect refrigerants until their chlorine atoms reached the stratosphere and catalytically shredded ozone. The Montreal Protocol (phase-out) is the standard environmental answer.

- **DDT (p,p'-dichlorodiphenyltrichloroethane):** the WWII miracle insecticide against malaria and typhus; its **stability** — the same property that made it durable — let it biomagnify up food chains, thinning eagle shells. Banned in many countries; resistance spread. Chemistry's most-quoted lesson in unintended consequences.

Key Reactions & Rules: One-Stop Revision

ROUTES, MECHANISMS, ORDERS

Rule / route	Symbols in plain words	When to reach for it
R–OH + $SOCl_2$ → R–Cl + SO_2 + HCl	gaseous by-products = pure product	The "cleanest conversion" question
Lucas reagent: HCl + $ZnCl_2$	3° instant, 2° minutes, 1° heat	Telling alcohol classes apart
Finkelstein: R–X + NaI, dry acetone	NaX precipitates, drives equilibrium	Making alkyl iodides
Swarts: AgF or Hg_2F_2	chloride → fluoride	Fluoroalkane preparation
Sandmeyer: ArN_2^+ + Cu_2X_2	diazonium → Ar–Cl/Br/CN, N_2 out	Aryl halides from anilines
Balz-Schiemann: $ArN_2^+BF_4^-$, heat	the Ar–F route	Aryl fluorides
S_N2: rate = k[RX][Nu^-]	one step, inversion, methyl > 1° > 2° > 3°	Strong Nu + aprotic solvent
S_N1: rate = k[RX]	carbocation, racemisation, 3° > 2° > 1°	Weak Nu + protic solvent
Saytzeff	elimination gives the more substituted alkene	Alcoholic KOH questions

Rule / route	Symbols in plain words	When to reach for it
R–X + Mg, dry ether → RMgX	Grignard; water destroys it	Every organomagnesium question
Wurtz / Fittig / Wurtz-Fittig	Na couples alkyl/aryl halides	Symmetrical hydrocarbon building
Dow process: 623 K, 300 atm	Ar–Cl + NaOH → phenol	Proof of aryl-halide inertness
o,p-directing but deactivating	resonance donates, -I withdraws	Halogen substitution on rings
$CHCl_3 \rightarrow COCl_2$ (air/light)	phosgene — store dark, full bottle	Chloroform storage questions
Freons destroy ozone; DDT biomagnifies	Montreal Protocol; banned insecticide	Environmental one-markers

Exceptions & Traps

THE TRAPS THIS CHAPTER SETS — READ BEFORE THE MOCK

The rule you assume	The exception / twist	The trap in the question
More electronegative halogen = stronger dipole	CH_3Cl beats CH_3F — bond LENGTH shrinks the lever	The dipole-moment comparison question with F vs Cl
S_N1 and S_N2 orders match	Exactly opposite: 3° wins S_N1, methyl wins S_N2	Every "which reacts faster" question — first read the mechanism asked
Racemisation = 50:50	S_N1 gives more inversion than retention	The departing X^- partially shields its face — not a perfect pair
Polarity makes haloalkanes water-soluble	They are water-insoluble — no H-bonding	"Which dissolves in water" options with alcohol/halide pairs
Halogen on benzene behaves like on alkane	Aryl halides refuse S_N — resonance + sp^2 + geometry	Chlorobenzene hydrolyses with NaOH easily — false; Dow conditions required

The rule you assume	The exception / twist	The trap in the question
Para isomer boils highest	p- isomer MELTS highest (packing), b.p. tracks branching	o-/m-/p-dichlorobenzene melting-point ordering
KCN and AgCN give the same product	KCN → R–CN (C-attack); AgCN → R–NC (N-attack)	The ambidentate nucleophile pairing question
Iodoform is an anaesthetic like chloroform	CHI_3 is the yellow antiseptic; $CHCl_3$ was the anaesthetic	Matching compounds to uses in one-mark grids

The Whole Chapter in One Picture

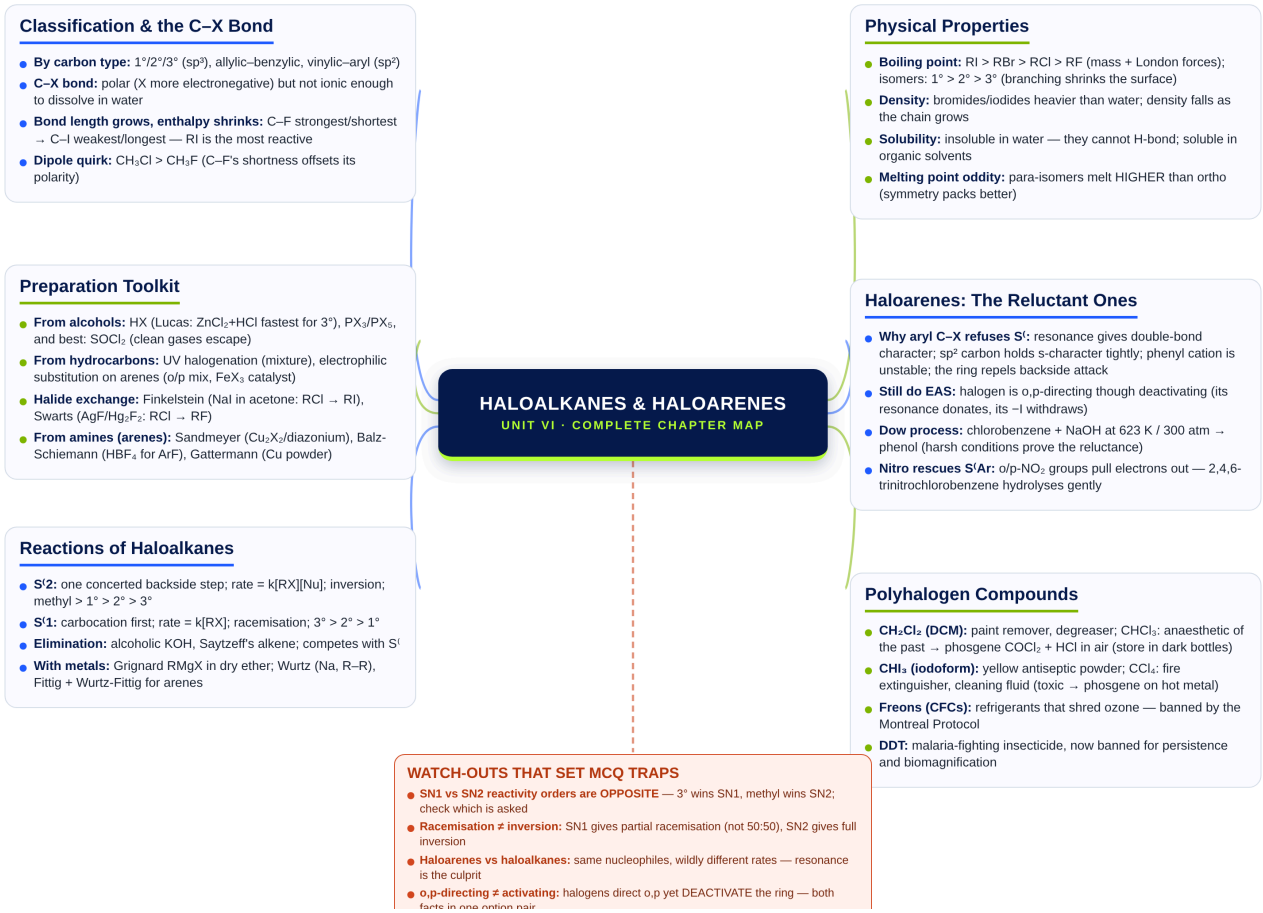


Figure 2: Chapter mindmap: classification and the C–X bond, preparation toolkit, physical trends, S_N1/S_N2 chemistry, the reluctance of haloarenes, and polyhalogen compounds — with the traps that catch most students.

Practice Questions (Exam Pattern)

Ten questions in the CUET UG style. Attempt all of them — the paper gives +5 for each correct answer and takes nothing away for a wrong one.

Q1. Which alkyl halide reacts fastest by the S_N2 mechanism?

(A) tert-butyl bromide
(B) isopropyl bromide
(C) ethyl bromide
(D) methyl bromide

Q2. A chiral alkyl halide undergoes hydrolysis with complete inversion of configuration. The mechanism is:

(A) S_N1 with racemisation
(B) S_N2 with backside attack
(C) E1 elimination
(D) S_N1 with retention

Q3. The order of reactivity of alkyl halides toward S_N1 reactions is:

(A) $1^\circ > 2^\circ > 3^\circ$
(B) $3^\circ > 2^\circ > 1^\circ$
(C) $2^\circ > 1^\circ > 3^\circ$
(D) all equal

Q4. The cleanest laboratory preparation of an alkyl chloride from its alcohol uses:

(A) PCl_5
(B) HCl with $ZnCl_2$
(C) $SOCl_2$
(D) Cl_2 under UV

Q5. Aryl halides are less reactive toward nucleophilic substitution because of all of the following EXCEPT:

(A) resonance gives the C–X bond partial double-bond character
(B) the sp^2 carbon holds the bond with more s-character
(C) phenyl cation is very unstable
(D) the halogen is less electronegative than in alkyl halides

Q6. In the Finkelstein reaction, the reagent and solvent are:

(A) AgF in water
(B) NaI in dry acetone
(C) NaBr in ethanol
(D) Cl_2 in CCl_4

Q7. Treating 2-bromobutane with hot alcoholic KOH gives mainly:

(A) but-1-ene
(B) but-2-ene (major, Saytzeff)
(C) butan-2-ol
(D) butane

Q8. The dipole moment of CH_3Cl is larger than that of CH_3F because:

(A) chlorine is more electronegative than fluorine
(B) the C–Cl bond is longer, and dipole = charge × distance
(C) fluorine repels the methyl group
(D) CH_3F is symmetrical but CH_3Cl is not

Q9. Chloroform is stored in dark bottles filled to the brim because:

(A) it evaporates rapidly
(B) light and air convert it to poisonous phosgene
(C) it reacts with glass
(D) it becomes more anaesthetic in the dark

Q10. The major environmental objection to freons (CFCs) is that they:

(A) cause acid rain
(B) deplete the ozone layer
(C) produce photochemical smog
(D) raise water temperature

Detailed Solutions

Sol. Q1. S_N2 is a backside attack — steric access decides. Methyl bromide exposes its carbon completely; tert-butyl blocks all four sides. Order: $CH_3X > 1^\circ > 2^\circ \gg 3^\circ$.

Answer: (D)

Sol. Q2. Complete inversion is the fingerprint of the concerted backside strike — one step, no planar intermediate. S_N1 gives a planar carbocation and therefore racemisation instead.

Answer: (B)

Sol. Q3. S_N1's slow step makes a carbocation, so stability rules: tertiary (hyperconjugation, inductive help) > secondary > primary. The exact mirror image of S_N2.

Answer: (B)

Sol. Q4. Thionyl chloride wins because both by-products — SO_2 and HCl — are gases that escape, leaving the alkyl chloride pure. PCl_5 leaves phosphorus residues; Lucas reagent is a classification test, not a prep.

Answer: (C)

Sol. Q5. Electronegativity of the halogen is irrelevant to aryl inertness. The real reasons: resonance double-bond character, sp^2 s-character, the unstable phenyl cation, and π-cloud repulsion of backside attack.

Answer: (D)

Sol. Q6. Finkelstein = NaI in dry acetone. The precipitating NaCl/NaBr drags the equilibrium to the iodide. AgF in the Swarts reaction makes fluorides instead.

Answer: (B)

Sol. Q7. Alcoholic KOH is basic — elimination. Saytzeff's rule favours the more substituted, thermodynamically comfortable alkene: but-2-ene (two alkyl groups on the double bond) beats but-1-ene.

Answer: (B)

Sol. Q8. Dipole moment = charge × separation. Chlorine is less electronegative than fluorine but sits farther out — the longer lever arm wins, so CH_3Cl edges CH_3F.

Answer: (B)

Sol. Q9. Oxygen and light oxidise chloroform to phosgene ($COCl_2$) plus HCl — a lethal gas. A dark, completely filled bottle excludes both; ethanol stabiliser helps too.

Answer: (B)

Sol. Q10. CFC chlorine atoms reach the stratosphere and catalytically convert ozone to oxygen — the hole over Antarctica. The Montreal Protocol phase-out is the standard remedy; acid rain and smog are different pollutant families.

Answer: (B)

Wrong-Answer Clinic — the one-line corrections

Dropped marks above? Find your slip here and rewrite the corrected rule in your own words.

If you answered...	Your slip was...	The corrected rule, in one line
Q1 option A	Applying 3° > 2° > 1° everywhere	That ladder belongs to S_N1; S_N2 reads methyl > 1° > 2° > 3° — always ask which mechanism
Q3 option A	Reusing the S_N2 order	Carbocation stability (S_N1): 3° > 2° > 1° — hyperconjugation feeds the cation
Q5 option D	Blaming electronegativity	Aryl inertness is structural (resonance/sp^2/geometry) — the halogen's electronegativity never enters
Q7 option A	Forgetting Saytzeff	Elimination picks the more substituted alkene — count the alkyl groups on the C=C
Q9 option A	Confusing storage with volatility	Dark, brim-full bottles keep out light AND air — the phosgene reaction needs both

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